

📘 AI/ML Internship – Day 29 Notes & Tasks

🟢 Module 5: Pandas (Sorting, Grouping & Data Analysis)

📌 Part 1: Why Data Analysis is Important?

👉 In AI/ML:

- Raw data alone is not useful
- We must analyze patterns and trends

👉 Example:

- Highest sales
 - Average marks
 - Department-wise salary
-

📌 Part 2: Theory Answers (IMPORTANT)

♦ 1. What is Sorting?

👉 Sorting means arranging data in:

- Ascending order
 - Descending order
-

♦ 2. Why Sorting is Important?

👉 Helps:

- Find highest/lowest values
 - Organize datasets
 - Improve analysis
-

♦ 3. What is Grouping?

👉 Grouping means combining rows with similar values.

👉 Example:

- Group employees by department
 - Group students by class
-

◆ 4. What is Aggregation?

👉 Aggregation means summarizing grouped data.

👉 Example:

- Sum
 - Average
 - Count
-

◆ 5. What is Data Analysis?

👉 Analyzing data to find patterns, insights, and useful information.

📌 Part 3: Create Dataset

```
import pandas as pd
```

```
data = {  
    "Name": ["Rahul", "Anu", "Arjun", "Meera"],  
    "Department": ["AI", "Web", "AI", "Web"],  
    "Salary": [50000, 40000, 60000, 45000]  
}
```

```
df = pd.DataFrame(data)
```

```
print(df)
```

📌 Part 4: Sorting Data

- ◆ Sort by Salary

```
print(df.sort_values("Salary"))
```

- ◆ Descending Order

```
print(df.sort_values("Salary", ascending=False))
```

- ✦ Part 5: Multiple Column Sorting

```
print(df.sort_values(["Department", "Salary"]))
```

- ✦ Part 6: Grouping Data

- ◆ Group by Department

```
group = df.groupby("Department")
```

```
print(group)
```

- ✦ Part 7: Group Aggregation

- ◆ Average Salary by Department

```
print(df.groupby("Department")["Salary"].mean())
```

- ◆ Total Salary

```
print(df.groupby("Department")["Salary"].sum())
```

- ◆ Employee Count

```
print(df.groupby("Department")["Name"].count())
```

- ✦ Part 8: Multiple Aggregations

```
print(  
    df.groupby("Department")["Salary"].agg(["mean", "max", "min"])  
)
```

📌 Part 9: Value Counts

```
print(df["Department"].value_counts())
```

👉 Counts repeated values.

📌 Part 10: Real-World Thinking

👉 In AI/Data Science:

- Sorting → ranking systems
- Grouping → business analysis
- Aggregation → statistical summaries

👉 Examples:

- Top-performing students
 - Department-wise analysis
 - Sales reports
-

📌 Part 11: Visualization of Grouping

Department: AI

Rahul → 50000

Arjun → 60000

Average Salary:

55000

Department: Web

Anu → 40000

Meera → 45000

Average Salary:

42500

Tasks for Day 29

Task 1: Theory (Write Answers)

1. What is sorting?
 2. Why sorting is important?
 3. What is grouping?
 4. What is aggregation?
 5. What is data analysis?
-

Task 2: Sorting Practice

1. Sort salary ascending
 2. Sort salary descending
 3. Sort by multiple columns
-

Task 3: Grouping Practice

1. Group by department
 2. Group by city
 3. Group by category
-

Task 4: Aggregation Practice

1. Find average salary

2. Find total salary
 3. Count employees
-

Task 5: Multiple Aggregation

1. Find mean
 2. Find max
 3. Find min
-

Task 6: Value Counts

1. Count departments
2. Count repeated values
3. Analyze category frequency